

Clinic Appoiment & Patient Record Management System

C# Windows Forms Project Report

1.INTRODUCTION

Manual record keeping is still common in many small private clinics. Patient details and appointments are often written in notebooks or managed through phone calls.

This leads to several issues such as:

- Double booking of appointments
- Loss of patient records
- Time wasted searching for information
- Poor daily schedule management

To overcome these problems, we developed a desktop-based Clinic Appointment & Patient Record Manager using C# Windows Forms, SQL Server, and LINQ-to-SQL.

This system helps clinic staff to:

- Register and log in securely
- Store patient details
- Create and manage appointments
- View daily appointment schedules
- Maintain records efficiently and digitally

2.OBJECTIVES

Main Objectives

Develop a simple Windows desktop application to manage clinic patients and appointments digitally.

Specific Objectives

- Implement secure user login and registration
- Maintain patient records (CRUD operations)
- Manage appointment scheduling
- Provide daily appointment reports

- Create a user-friendly interface

3. TECHNOLOGIES USED

1. Programming Language: C#
2. Framework: .NET Windows Forms
3. Database: SQL Server
4. Data Access: LINQ-to-SQL
5. IDE: Visual Studio

4. SYSTEM ARCHITECTURE

The system follows a 3-layer structure:

1. Presentation Layer

- Windows Forms (UI)
- Buttons, TextBoxes, DataGridViews

2. Business Logic Layer

- Validations
- Event handling
- Form interactions

3. Data Layer

- SQL Server database
- LINQ queries
- Tables: Users, Patients, Appointments

5. DATABASE DESIGN

5.1 User Table

Field	Type	Description
UserId	Int	Unique ID
Username	Varchar	Login name
Password	Varchar	User password
FullName	Varchar	Staff name
Role	Varchar	Admin/Receptionist/

		Doctor
Status	Varchar	Active/Inactive

Purpose:

- Store Login information.

5.2 Patients Table

Field	Type	Description
patientId	Int	Unique ID
Name	Varchar	patient name
Phone	Varchar	Contact number
Address	Varchar	Adress

Purpose:

- Store patient details.

5.3 Appointment Table

Field	Type	Description
AppointmentId	Int	Unique ID
PatientName	Varchar	Doctor name
DoctorName	Varchar	Contact number
AppointmentDate	date	Date
TimeSlot	varchar	TimeSlot
Status	varchar	Scheduled/Completed/ Cancelled

Purpose:

- Manage Clinic Schedules.

6.SYSTEM FUNCTIONALITIES

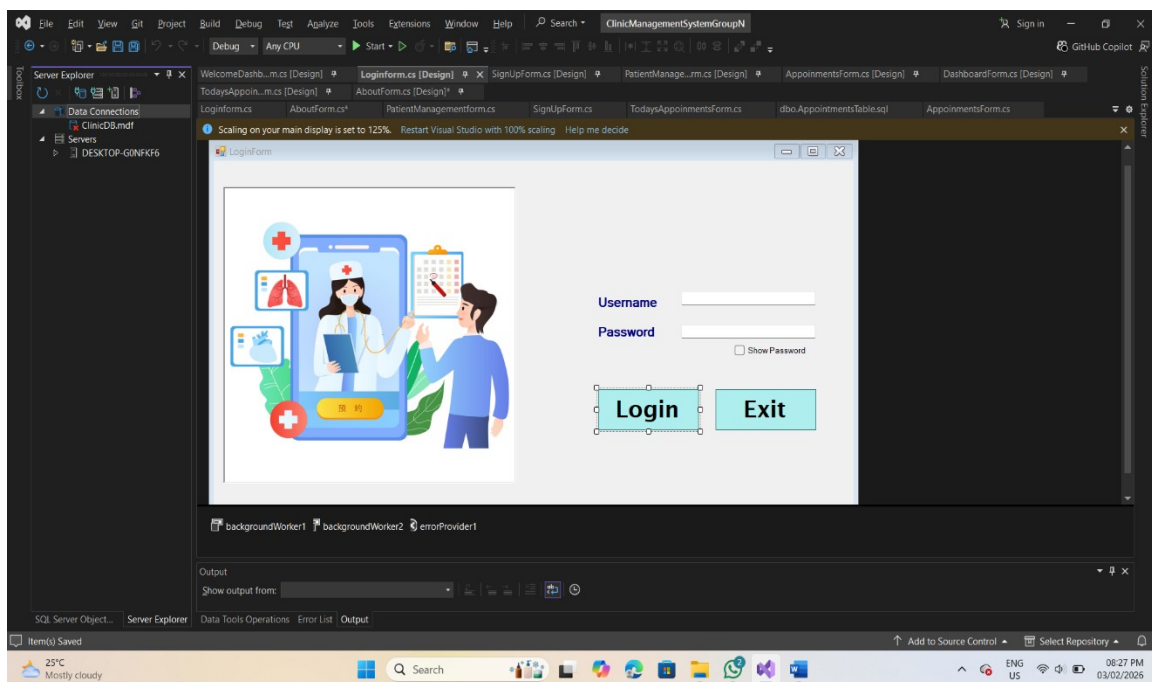
The system provides the following features:

- User Registration and Login
- Patient Management

- Appointment Scheduling
- Daily Report Generation
- Dashboard Navigation
- About & Help Information

7. Forms Description

7.1 Login & Sign Up Form



This is the first screen of the system. Users can either log in with existing credentials or register a new account.

Purpose

Allows users to create accounts and log in securely.

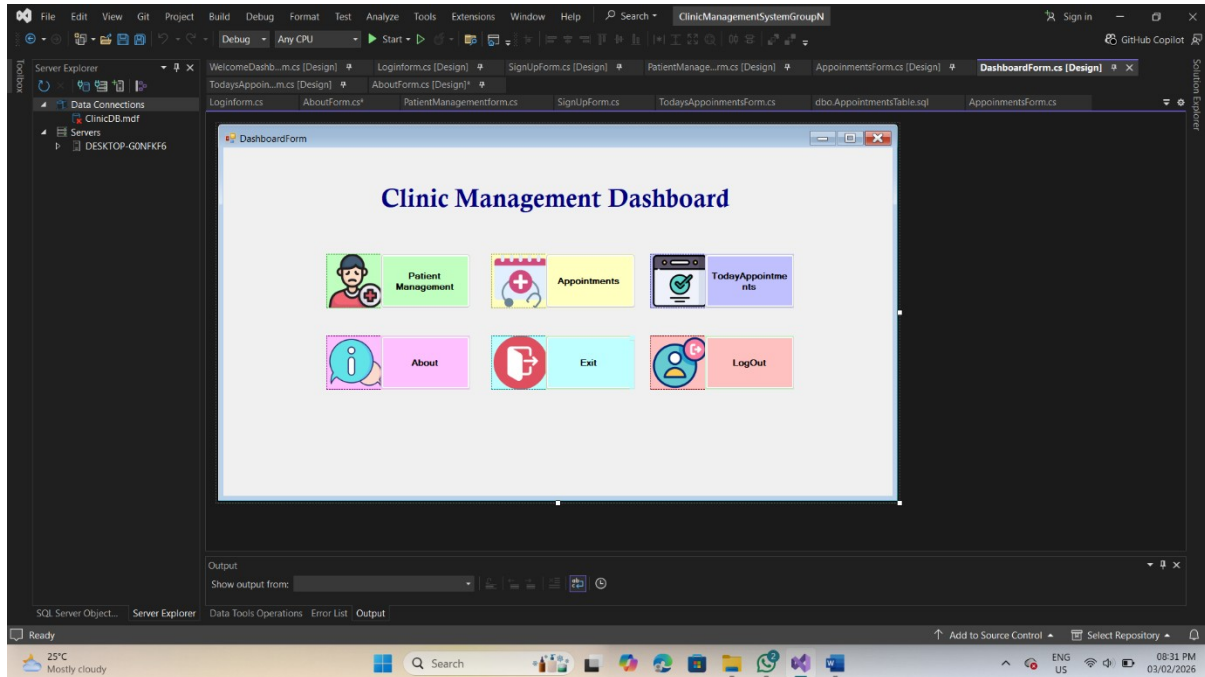
Controls Used

- TextBoxes (Username, Password)
- Buttons (Login, Register)
- ComboBox (Role)

Functions

- Validate input
- Insert new users
- Authenticate login credentials

7.2 Dashboard Form



The dashboard acts as the main navigation page after successful login. Users can access all system modules from here.

Purpose

Acts as the main navigation page after login.

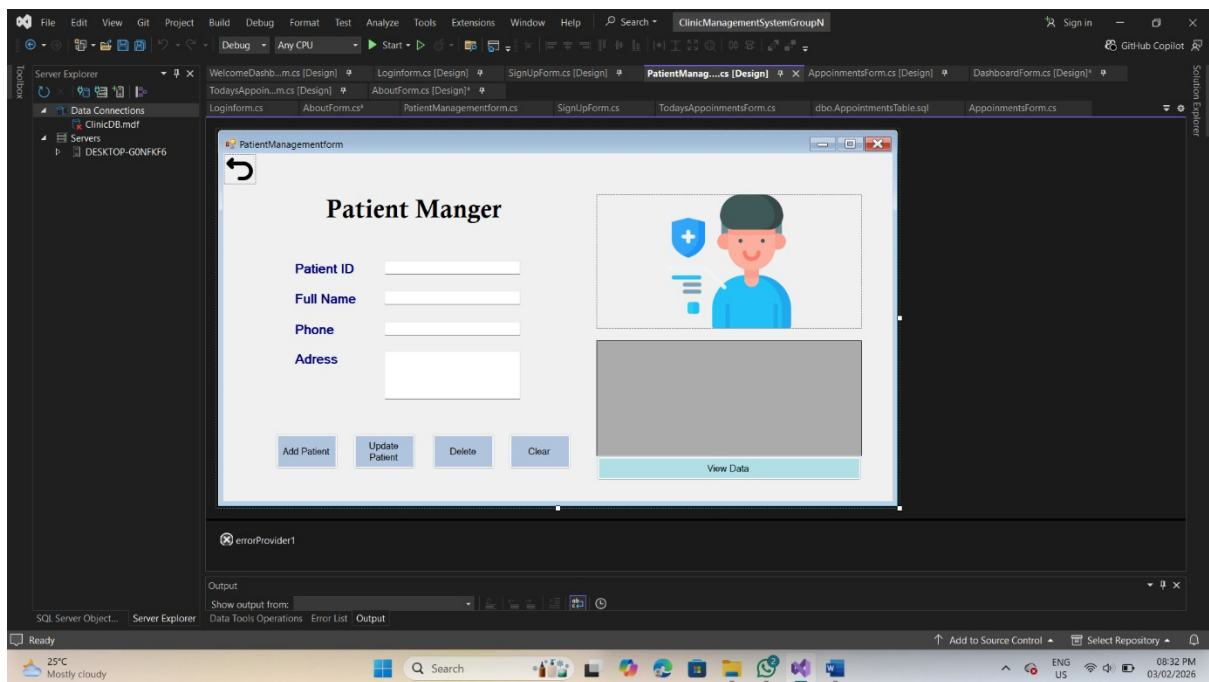
Features

- Open Patients Form
- Open Appointments Form
- View Today's Appointments
- About
- Logout/Exit

Note

- No database operations occur here.

7.3 Patient Management Form



This form is used to manage patient records. Staff can add, edit, delete, and search patient information.

Purpose

Manage patient information.

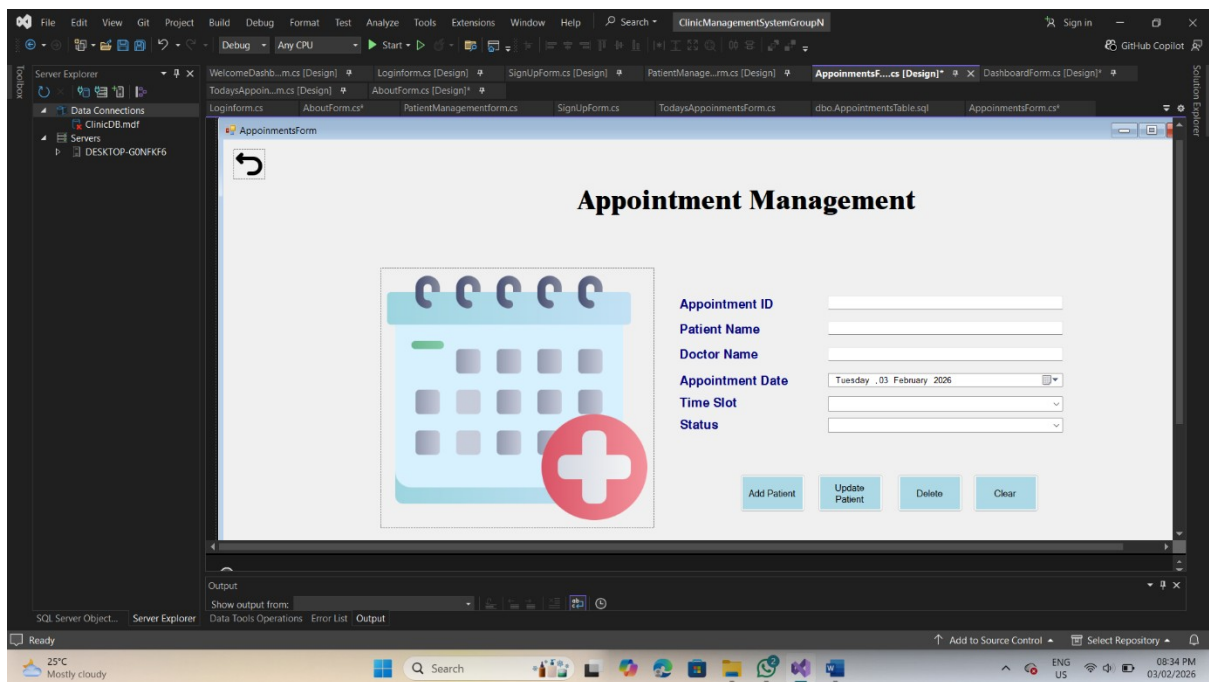
Functions

- Add new patient
- Update patient
- Delete patient
- Display patients in DataGridView

Operations

CRUD using LINQ queries.

7.4 Appointment Management Form



This form allows staff to schedule appointments between patients and doctors.

Purpose

Schedule and manage appointments.

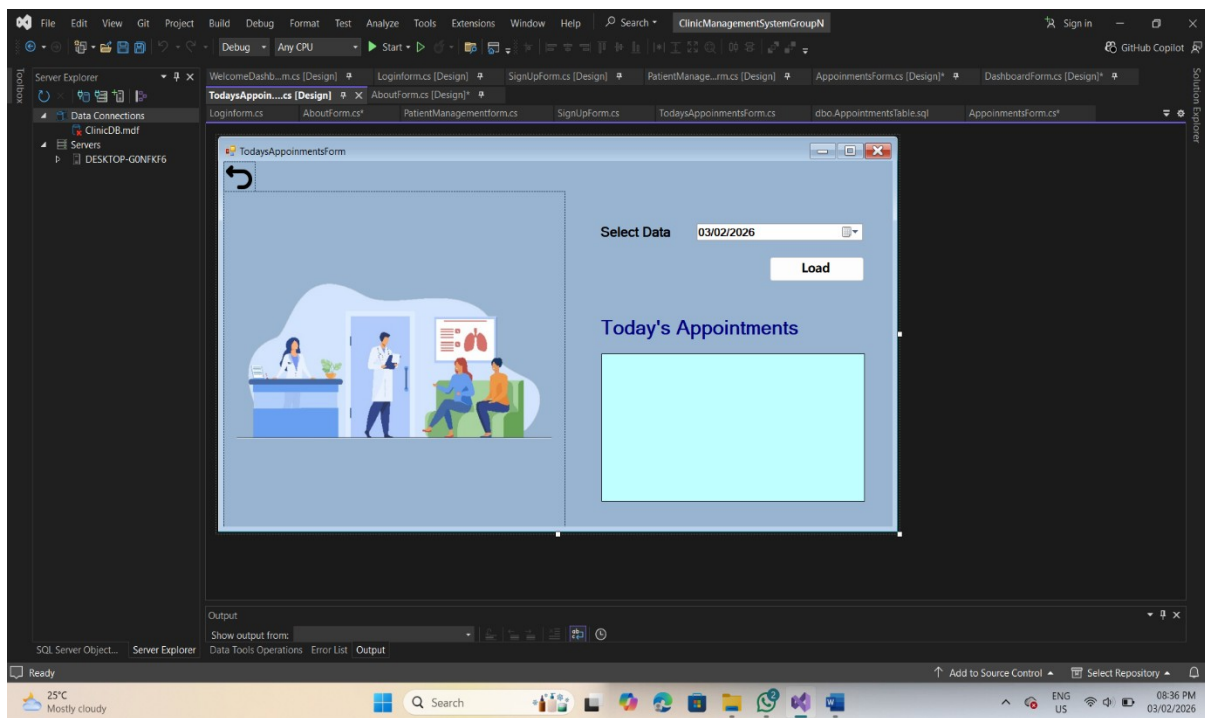
Functions

- Add appointment
- Modify appointment
- Delete appointment
- Select date and time slot
- Change status

Importance

Core functionality of the system.

7.5 Today's Appointments Report Form



This report screen displays all appointments for a selected date. It helps staff quickly view the day's schedule.

Purpose

Display appointments for a selected date.

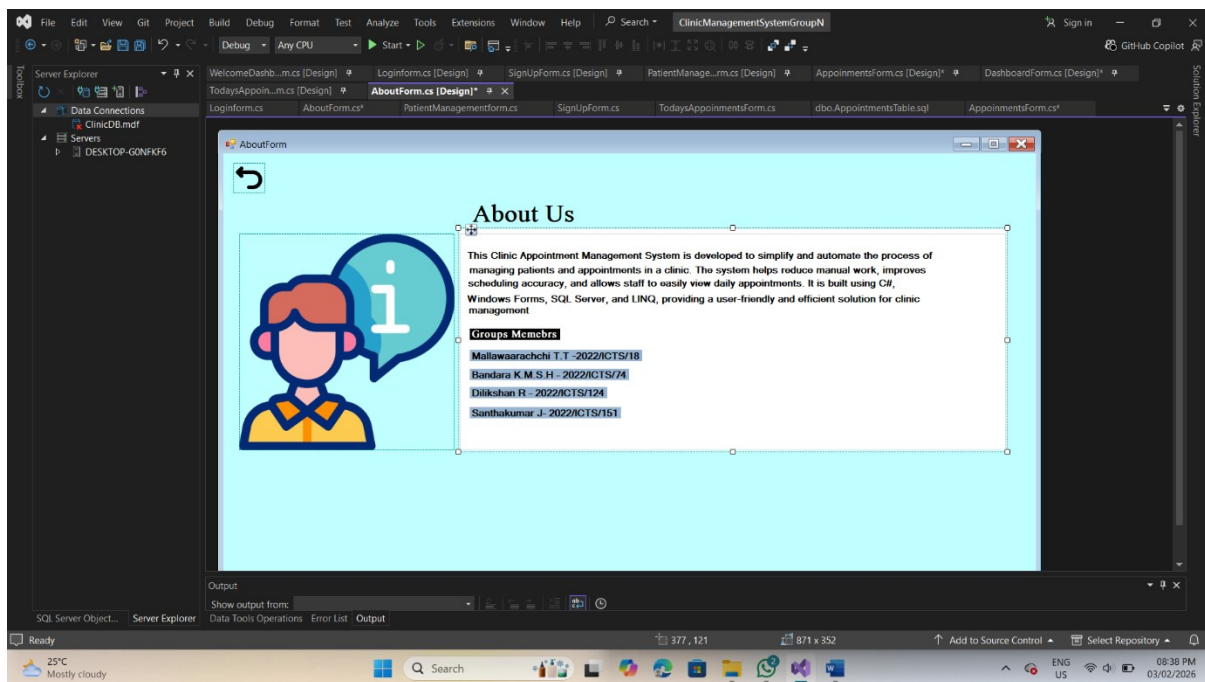
Functions

- Choose date
- Filter appointments
- Show schedule in grid

Benefit

Helps staff plan daily work easily.

7.6 About Form



This form provides information about the system and the development team.

Purpose

Provide project and team details.

Displays

- Project title
- Description
- Technologies used
- Team members
- Roles

Note

- Read-only information.

8. LINQ-to-SQL Usage

LINQ-to-SQL is used to:

- Insert data
- Update records
- Delete rows
- Filter search results
- Connect forms with database

Example Operations

- InsertOnSubmit()

- DeleteOnSubmit()
- SubmitChanges()
- Where()
- FirstOrDefault()

9. Advantages of the System

- ❖ Easy to use interface
- ❖ Fast data retrieval
- ❖ Reduces paperwork
- ❖ Prevents double booking
- ❖ Secure login
- ❖ Organized records

10. Limitations

- ❖ Desktop only (not web-based)
- ❖ Single clinic use
- ❖ No cloud backup

11. Future Enhancements

- ❖ Online booking system
- ❖ SMS notifications
- ❖ Multiple doctor management
- ❖ Billing system
- ❖ Cloud database
- ❖ Reports & analytics

12. Conclusion

The Clinic Appointment & Patient Record Manager successfully replaces manual clinic record keeping with a digital system. It improves efficiency, reduces errors, and saves time for staff.

By using C#, Windows Forms, SQL Server, and LINQ, we gained practical experience in:

- ❖ Desktop application development
- ❖ Database design
- ❖ CRUD operations

- ❖ Event-driven programming
- ❖ Team collaboration

This project demonstrates how software solutions can simplify real-world healthcare management tasks.

13. Team Members Details.....

- 1) 2022/ICTS/18 (MALLAWAARACHCHI.T.T)
- 2) 2022/ICTS/47 (BANDARA G.M.C.D)
- 3) 2022/ICTS/74 (BANDARA K.M.S.H)
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